

Module 4 in-class Coding (Java Swing Employee Management Program)



Java Swing Employee Management System Lab Assignment

Objective:

Design and implement a Java Swing-based Employee Management System that uses MariaDB for storing and managing employee records. The application should allow users to add, update, and delete employee records and display the current list of employees.

Prerequisites:

- Basic knowledge of Java programming.
- Understanding of Java Swing for building GUI applications.
- Familiarity with JDBC for database connectivity.
- MariaDB installed and accessible.

Tasks:

1. Setup MariaDB Database (15 points):

- Create a MariaDB database named `company_db`.
- Design a table named `employees` with columns: `id` (INT, AUTO_INCREMENT, PRIMARY KEY), and `name` (VARCHAR(255)).

2. Implement GUI (25 points):

- Create a Java Swing GUI with the following components:
 - JLabel for "Name"
 - JTextField for entering employee name
 - JButton for "Add"
 - JButton for "Update"
 - JButton for "Delete"

- JTextArea for displaying the list of employees (10 rows, 40 columns)

3. Add Employee (20 points):

- Implement the logic to add a new employee to the database when the "Add" button is clicked.
- Ensure that the employee name is not empty before adding.

4. Update Employee (20 points):

- Implement the logic to update an existing employee in the database when the "Update" button is clicked.
- Allow the user to select an employee from the displayed list and modify the name.

5. Delete Employee (20 points):

- Implement the logic to delete an existing employee from the database when the "Delete" button is clicked.
- Ensure that the user selects an employee from the displayed list before deleting.

6. Display Employees (15 points):

- Fetch and display the list of employees from the database when the application starts.
- Update the display after each add, update, or delete operation.

7. Skeleton code:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;

public class DatabaseSetup {

    public static void main(String[] args) {
        try {
            // Connect to MariaDB
            Connection connection = DriverManager.getConnection("jdbc:mariadb://localhost:3306/", "username", "password");

            // Create a new database
            Statement statement = connection.createStatement();
            statement.executeUpdate("CREATE DATABASE IF NOT EXISTS company_db");

            // Use the created database
            statement.execute("USE company_db");

            // Create employees table
            String createTableQuery = "CREATE TABLE IF NOT EXISTS employees ("
                + "id INT AUTO_INCREMENT PRIMARY KEY,"
                + "name VARCHAR(255),"
                + "position VARCHAR(255),"
                + "salary DOUBLE)";
            statement.executeUpdate(createTableQuery);

            System.out.println("Database and Table created successfully.");

            // Close connections
            statement.close();
            connection.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.util.ArrayList;
import java.util.List;

public class EmployeeManagementApp {

    public static void main(String[] args) {
        JFrame frame = new JFrame("Employee Management System");
        frame.setSize(600, 400);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JPanel panel = new JPanel();
        frame.add(panel);
        placeComponents(panel);

        frame.setVisible(true);
    }

    private static void placeComponents(JPanel panel) {
        panel.setLayout(new FlowLayout());

        JLabel nameLabel = new JLabel("Name:");
        panel.add(nameLabel);

        JTextField nameText = new JTextField(20);
        panel.add(nameText);

        JButton addButton = new JButton("Add");
        panel.add(addButton);

        JButton updateButton = new JButton("Update");
        panel.add(updateButton);

        JButton deleteButton = new JButton("Delete");
        panel.add(deleteButton);

        JTextArea displayArea = new JTextArea(10, 40);
        panel.add(displayArea);
    }
}
```

```
// Action listeners for buttons
addButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        // Add record logic here
        displayArea.setText("Add button clicked");
    }
});

updateButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        // Update record logic here
        displayArea.setText("Update button clicked");
    }
});

deleteButton.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        // Delete record logic here
        displayArea.setText("Delete button clicked");
    }
});
}
```

Grading Rubric:

- **MariaDB Database Setup:**

- Company_db creation: 5 points
- Employees table creation: 10 points

- **GUI Implementation:**

- JLabel, JTextField, JButtons, and JTextArea: 10 points

- Proper layout (using FlowLayout): 15 points
- **Add Employee:**
 - Logic to add a new employee: 20 points
 - Proper handling of empty name: 5 points
- **Update Employee:**
 - Logic to update an employee: 20 points
 - Proper handling of empty name and selection: 5 points
- **Delete Employee:**
 - Logic to delete an employee: 20 points
 - Proper handling of selection: 5 points
- **Display Employees:**
 - Fetch and display employees from the database: 15 points
- **Code Structure and Comments:**
 - Clear and organized code: 10 points
 - Comments explaining key sections: 5 points

Submission Guidelines:

- Submit a zip file containing the Java source code files (.java).
- Include a README file with any necessary instructions, especially MariaDB setup details.
- Ensure code readability and follow best practices.

Note: Adjust the point distribution based on the complexity of your class requirements and your specific grading criteria.

Points 100

Submitting a file upload

Due	For	Available from	Until
Oct 5	Everyone	-	Oct 22 at 11:59pm